1. Implementation Topic

This tender applies to Implementation Topic #2: Connect sensors to SensorThings API server.

2. Motivation

Across the Netherlands, municipalities operate thousands of smart waste containers with fill-level sensors, RFID identification, and weighing systems on collection vehicles. Today this sensor data is locked inside proprietary platforms, inaccessible for cross-domain analysis, digital twin applications, or integration with broader spatial data infrastructures.

Geonovum has explicitly invited parties within the DTaaS (Digital Twin as a Service) eco system to expose their sensors through the OGC SensorThings API. Clappform B.V. (lead) with SULO as research partner steps forward as a DTaaS-aligned bidder offering exactly that: real, operational municipal waste sensor streams connected to a SensorThings API server, in a sensor domain that is distinct from and complementary to the wastewater use case provided in this testbed.

SULO is Europe's largest provider of smart waste solutions, operating sensor ecosystems across 3,000+ vehicles and 10,000,000+ RFID transponders. Clappform is a Dutch data integration platform that already runs in production at the Afvalstoffendienst 's-Hertogenbosch (serving 's-Hertogenbosch and six surrounding municipalities), connecting ERP, sensor platforms, weighing systems, and reporting dashboards into a single integrated data layer. Together we bring sensors, domain expertise, and integration engineering to the testbed.

The municipal customer need is clear: connect sensor data to the digital twin in a robust way and through a single standard. Meeting that need requires close collaboration with the municipalities in the region, and we see the same need at scale in cities such as Alkmaar and Almere, where we aim to contribute within their DTaaS projects. Linking the digital twin to environmental and waste-collection data also supports municipal sustainability targets: by feeding real sensor observations back into the twin, different operational and policy scenarios (collection routing, container placement, emissions) can be modelled and compared more accurately, turning the digital twin from a static representation into a feedback-driven decision tool.

3. Plan of Approach

3.1 Approach

We connect SULO's fill-level sensors from operational municipal waste containers to the central SensorThings API server, using Clappform's data platform as the intermediary translation layer that transforms SULO's proprietary sensor data into OGC-compliant SensorThings API observations. Primary stream: fill-level (continuous time-series, ideal STA fit). Secondary streams: RFID collection events and vehicle GPS, demonstrating that the same model accommodates both static (container) and mobile (vehicle) platforms.

3.2 Architecture

- **Source layer (SULO):** SmartSULO platform exposes sensor data through existing APIs; container fill-level via IoT connectivity, vehicle data via on-board hardware (SmartSULO ID).
- **Translation layer (Clappform):** connects to SULO's APIs, transforms proprietary data into OGC OMS-compliant structures, pushes observations to the STA server. Handles mapping, validation, quality checks, and synchronization.
- **Target layer:** preferred target is the central FROST server hosted by the Topic #1 contractor; we will align our OMS model with theirs. Fallback: if no central server is available at testbed start, Clappform deploys a dedicated FROST-Server (Docker, in our Kubernetes environment) and migrates observations to the central server once available, guaranteeing deliverables are not blocked by external dependencies.

3.3 OMS Data Model

Thing = waste container (georeferenced, EPSG:4326) or collection vehicle. Sensor = fill-level device (ultrasonic/radar) with manufacturer/model metadata. ObservedProperty = fill level (%), with secondary properties for temperature and battery. Datastream = per-container time-series with defined observation type and unit. Observation = individual reading with phenomenonTime, resultTime, result. FeatureOfInterest = container location as GeoJSON Point. Same model applies to vehicle datastreams (GPS, collection events).

3.4 Phasing

Phase	Period	Activities
1. Setup & Align	May 2026	Align with Topic #1 contractor on data model and server config. Define OMS mapping for waste container sensors. Set up Clappform integration pipeline. Establish API access to SULO's environment.
2. Build & Connect	June 2026	Implement translation layer in Clappform. Connect fill-level sensors first, then RFID + vehicle GPS. Validate by comparing STA output against SULO's native platform. Run data quality checks.
3. Document & Demo	July 2026	Reproducibility guide. Public demonstration. Document lessons learned, obstacles, recommendations for STA adoption in the waste sector. Final report.

3.5 Validation, Reproducibility, Use-Case Engagement

Validation: compare observations on the STA server against SULO's native SmartSULO platform (values, timestamps, metadata); automated data quality checks in Clappform (missing observations, out-of-range values, sync delays); STA query verification (filtering by time, location, observed property); historical-data stress test at municipal scale (hundreds of containers, multiple streams).

Reproducibility and vendor-agnostic design: we deliberately design the OGC SensorThings API translation layer so that it works today with SULO's sensors but extends naturally to other waste-container sensor vendors in the future. The translation layer abstracts vendor-specific protocols (HTTP/REST, MQTT, modbus, proprietary cloud APIs) into a single OMS-compliant data flow into the STA server, and the reproducibility guide includes a step-by-step worked example covering OMS modelling for waste-container sensors, configuring a translation layer between any proprietary sensor API and STA, common pitfalls, and a reference implementation that other vendors and municipalities can adopt. The demand for connecting waste-container sensor data to municipal digital twins is large and growing, driven by societal and sustainability goals (collection efficiency, lower emissions, fewer collection kilometres, cleaner public space), and a vendor-agnostic standard is the only way to scale that demand without locking municipalities into a single supplier. Published as open documentation under CC-BY 4.0.

Engagement with the Brabantse Delta wastewater use case: (1) where possible, push observations to the same central FROST server so wastewater pressure/flow sensors and waste-container fill-level sensors coexist in one OMS-modelled environment, a realistic scenario for any municipality combining sensor domains; (2) share OMS modelling decisions in the open testbed sessions, comparing how time-series data from very different physical processes maps onto the same standard; (3) include a wastewater worked example alongside the waste-container example in the reproducibility guide.

Beyond this testbed, the same translation layer can be plugged into existing Dutch municipal digital twin environments (e.g. Alkmaar, Almere) to standardise sensor ingestion via OGC STA, replacing point-to-point connectors and aligning with Geonovum's DTaaS direction.

4. Partners, References and Performers

4.1 Research Partners

Clappform B.V. (lead) is a Dutch data management and integration platform headquartered in the Netherlands. The platform connects disparate operational systems, applies data quality and validation, and exposes integrated data through APIs and applications. Clappform is active in the Dutch public sector (waste management, housing, municipalities and provinces) and runs production deployments at clients including the Afvalstoffendienst 's-Hertogenbosch, Provincie Zuid-Holland, and the Municipality of Zaanstad. For this testbed, Clappform provides the integration engineering, the OGC SensorThings API translation layer, GIS modelling, and project delivery.

SULO Group (research partner) is Europe's largest provider of sustainable waste-management solutions, with a deployed footprint of 3,000+ collection vehicles and 10 million+ RFID transponders worldwide. For this testbed, SULO provides the operational sensors, SmartSULO API access, and domain expertise.

Gemeente 's-Hertogenbosch (municipality) is the launching customer for waste data and digital twin integration in this testbed. The municipality, through the Afvalstoffendienst, operates waste collection for 's-Hertogenbosch and six surrounding municipalities and provides the operational context, real-world requirements, and digital twin integration target. Client contact: Herman Kasdiran (h.kasdiran@s-hertogenbosch.nl, +31 6 112 46 804).

Clappform and SULO already collaborate on Dutch waste-management data integration and analytics for municipalities (for example, illegal-dumping detection dashboards), giving the consortium a working baseline of integration patterns, data semantics, and trust before testbed start.

4.2 References

- **Clappform, Afvalstoffendienst 's-Hertogenbosch:** data platform delivered for the Afvalstoffendienst and the six surrounding municipalities in the region (regional collaboration with Vught, Oisterwijk, Heusden, Bernheze, Altena, Boxtel), integrating ERP, sensor platforms, weighing systems, and reporting dashboards into a single integrated data layer. Client contact for waste data and digital twin integration: Herman Kasdiran, Gemeente 's-Hertogenbosch (h.kasdiran@s-hertogenbosch.nl, +31 6 112 46 804).
- **Clappform, Dutch public-sector data integration:** active engagements with Provincie Zuid-Holland and the Municipality of Zaanstad on housing-sector data integration, including municipal data products under the Omgevingswet framework.
- **SULO, SmartSULO deployments:** operational at municipal authorities including Paris, Grand Besançon, and multiple SMICTOM inter-municipal authorities in France.
- **Existing Clappform/SULO collaboration:** data integration and analytics for Dutch municipalities, including illegal-dumping detection dashboards built on combined sensor and operational data.
- **Code and documentation:** Clappform Python API wrapper on PyPI (*https://pypi.org/project/clappform/*) and the OGC Digital Twin work on GitHub (*https://github.com/ClappFormOrg/OGC-Digital-twin*). The GitHub repository is private; access can be granted to the Geonovum review team on request.

4.3 Performers of the Research

Name	Role	Background and relevant experience
Diego Tolen	Solution Architect (Clappform)	Founder of Clappform. Background in Computer Science. Former Big Data team at KPMG, with substantial experience in API design and data standardisation. Owns the technical architecture and API integration design for the testbed; data integration for the Dutch public sector and waste-management domain.
Tom Greiffioen	Project Management (Clappform)	Project delivery, planning, and stakeholder coordination. Single point of contact for the Geonovum bi-weekly meetings; coordinates sprints across the Clappform team and the SULO partner.

Name	Role	Background and relevant experience
Bowen Harkema	OGC API / Software Engineer (Clappform)	6 years at Clappform. Involved in Testbed 2.0. Currently works on the digital twin infrastructure of the municipalities of Almere and Alkmaar. Implements the OGC SensorThings API translation layer, OMS data-model implementation, and FROST-Server configuration.
Yashvir Jhingur	OGC API / Software Engineer (Clappform)	6 years at Clappform. Involved in Testbed 2.0. Currently works on the digital twin infrastructure of the municipalities of Almere and Alkmaar. Implements the OGC SensorThings API translation layer, SULO API integration, and validation tooling.
Wybren Terpstra	GIS Specialist (Clappform)	Involved in the GIS layer and GeoServer setup for the municipality of Alkmaar. Geospatial modelling for FeatureOfInterest and container locations (EPSG:4326), spatial query design, and validation of geo-referenced sensor metadata.
Vincent Esajas	Domain Expert (SULO)	SULO contact for the consortium (vincent.esajas@sulo.com). Provides API access to the SmartSULO platform, sensor specifications, and operational data; coordinates OMS data-model alignment with SULO's product team.

Extended CVs and references for individual team members can be provided on request.

5. In-Kind Investment

- **SULO:** operational sensor data access, SmartSULO API access, domain expertise and consultation. Estimated in-kind value €8,000–12,000.
- **Clappform:** platform licensing and hosting, engineering for the translation layer and data quality checks, participation in testbed sessions and knowledge sharing. Estimated in-kind value €10,000–15,000.

Total project investment (testbed budget + in-kind): approximately €23,000–32,000.

6. Publication Agreement

Both Clappform B.V. and SULO Group agree to publication of all research results and deliverables under Creative Commons Attribution 4.0 International (CC-BY 4.0). Software developed during the testbed will be released under an open-source license (MIT or Apache 2.0). The translation layer configuration and documentation will be added to the Geonovum GitHub repository. The integration pipeline, connected sensors, and STA endpoint remain operational and publicly accessible for at least six months after the testbed concludes.